

Scaling Window For Adaptable 2-Colorability

Thomas Snow

November 5, 2025

Adapting an Edge Coloring

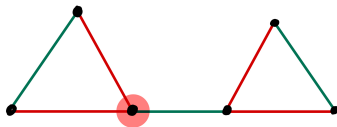
Given an edge coloring $c : E \rightarrow \{1, \dots, k\}$ for a graph $G = (V, E)$, we say (G, c) is **adaptable k -colorable** if there exists a vertex coloring $\sigma : V \rightarrow \{1, \dots, k\}$ such that for every edge uv either

$$c(u) \neq \sigma(u) \quad \text{or} \quad c(v) \neq \sigma(v)$$

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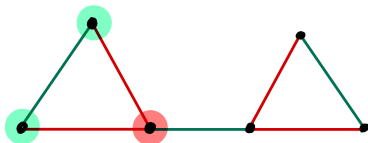
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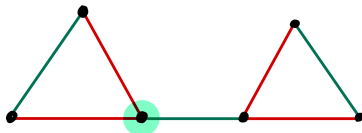
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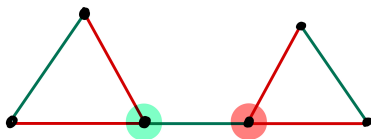
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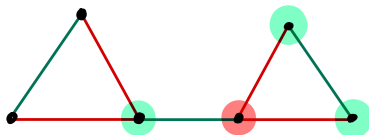
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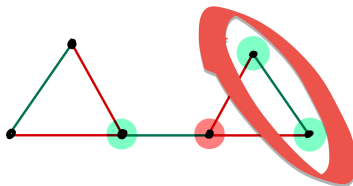
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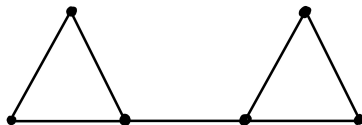
A graph $G = (V, E)$ is **adaptable k -colorable** if for **every** edge coloring $c : E \rightarrow \{1, \dots, k\}$, (G, c) is adaptably k -colorable.

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$$\chi_a(\uparrow) > 2$$

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Lemma

For any graph G , $\chi_a(G) \leq \chi(G)$

Properties of χ_a

Lemma (Hell, Zhu '08)

For $S \subseteq E$, if $k := \chi(G \setminus S) > |S|$ then $\chi_a(G) \leq \chi(G \setminus S)$

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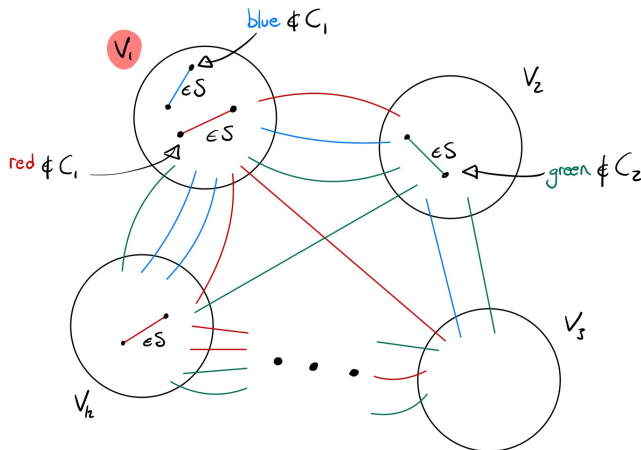
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- For $e \in S$ contained in some V_i , say e forbids $c(e)$ from V_i
- C_i denotes the set of available colors for V_i

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Proof Continued

Distinct Representatives for $C_1, \dots, C_k$



(G, c) is adaptably k -colorable.

Otherwise,

- Hall's Theorem $\implies$ There exists $A \subseteq \{C_1, \dots, C_k\}$ s.t.

$$I := |A| > \left| \bigcup_{C \in A} C \right|$$

- at least $I(k - 1 + 1) \geq k$ forbidden colors
- Contradicting $|S| < k$



Characterizing $\chi_a(G) = 2$

Theorem (Theorem 2.1 in [Hell, Zhu '08])

The following statements are equivalent for a connected graph G ,

- 1 G is adaptably 2-colorable.
- 2 G does not have an odd bicycle or an odd edge- K_4 .
- 3 There is an edge e such that $G - e$ is bipartite.

$\implies$ Same Threshold as the giant component!

Random Graphs

$\mathcal{G}(n, p, k)$

- Sample $G \sim \mathcal{G}(n, p)$ from the Erdős-Rényi model.
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Question

For what $p = p(n)$ is $\mathcal{G}(n, p, 2)$ adaptably 2-colorable with probability $1 - o(1)$?

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Theorem

For any constant $\epsilon > 0$ we have that,

- 1 $\mathcal{G}(n, p = \frac{2-\epsilon}{n}, 2)$ is w.h.p adaptably 2-colorable.
- 2 $\mathcal{G}(n, p = \frac{2+\epsilon}{n}, 2)$ is w.h.p not adaptably 2-colorable.

When is (G, c) Not 2-Colorable?

Question

For what colorings $c : E \rightarrow \{\text{red}, \text{blue}\}$ is (G, c) not adaptably 2-colorable?

Consider an alternating path P

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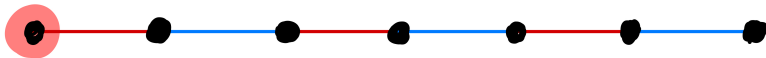


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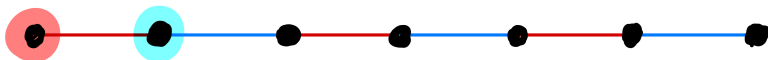


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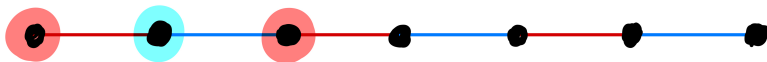


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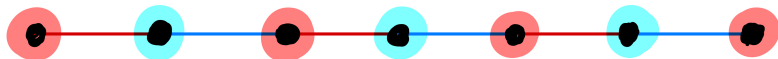


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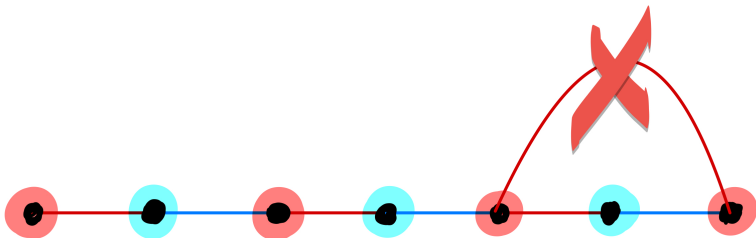


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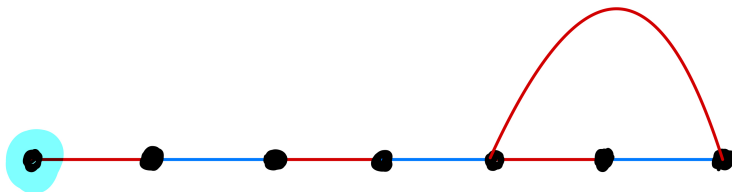


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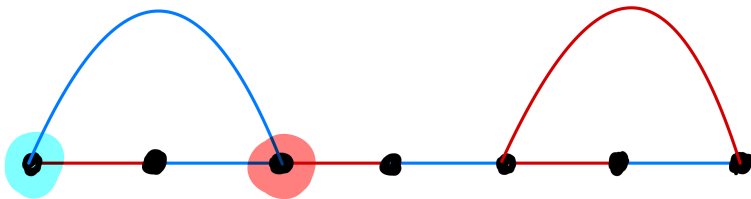


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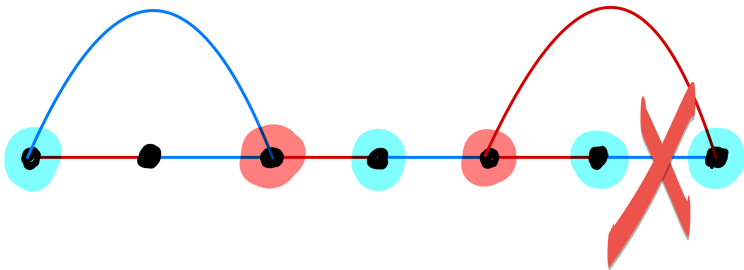


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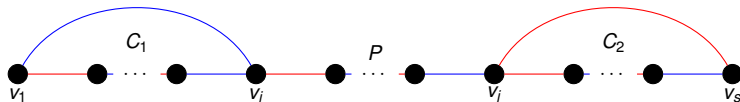
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Alternating Bicycles

Alternating Bicycle

- Alternating path $P = v_1 v_2 \cdots v_s$
- Edges $v_1 v_i$ and $v_j v_s$ for $i, j \in \{1, \dots, s\}$ (We don't require $i \leq j$).
- $c(v_1 v_i) \neq c(v_1 v_2)$ and $c(v_j v_s) \neq c(v_s v_{s-1})$



- If $i \leq j$ call it **proper**.
- If cycles C_1 and C_2 are odd, call it **odd**.

When is (G, c) Not 2-Colorable?

We already showed:

Lemma

If (G, c) contains a proper alternating odd bicycle, then (G, c) is not adaptably 2-colorable.

We now show:

Lemma

If G is connected and (G, c) is not adaptably 2-colorable, then (G, c) contains an alternating bicycle.

When is (G, c) Not 2-Colorable?

Lemma

If G is connected and (G, c) is not adaptably 2-colorable, then (G, c) contains an alternating bicycle.

Proof.

- Assume (G, c) is not adaptably 2-colorable.

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- Assume (G, c) is not adaptably 2-colorable.
- Let σ be a vertex coloring with minimal "violated" edges.



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- Let $e = uv$ be a violated edge under σ

$$c(e) = \sigma(u) = \sigma(v) = \text{red}$$

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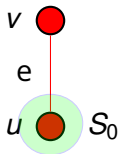


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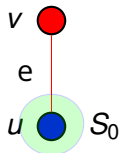
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- Consider swapping the color of u .

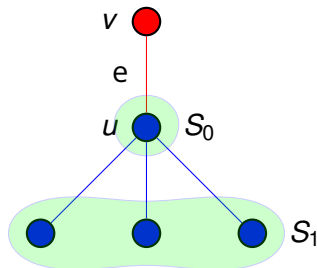
Proof (Continued)



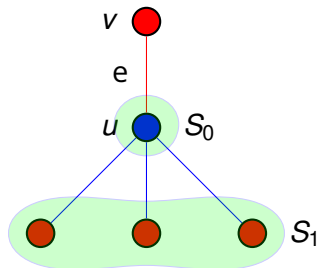
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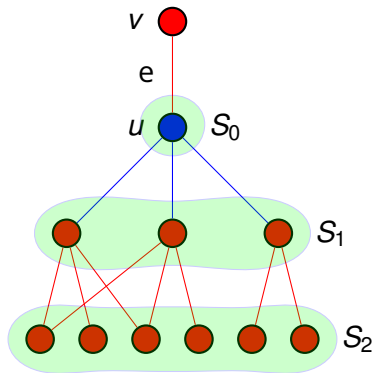
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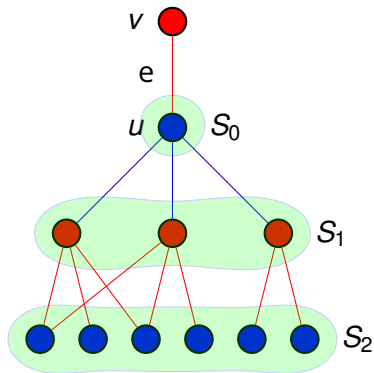
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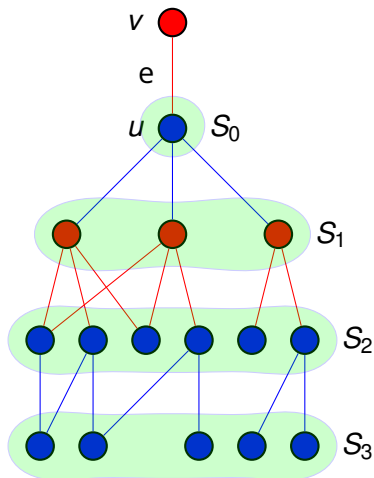
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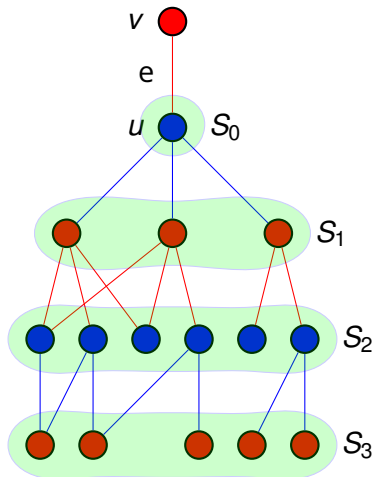
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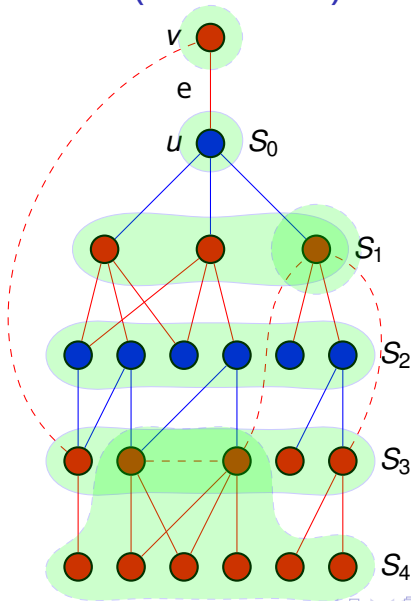
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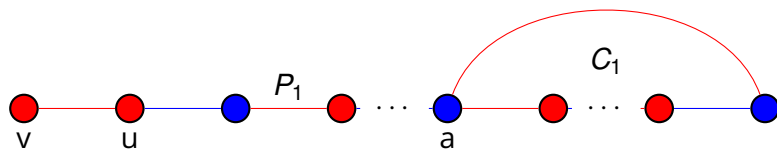
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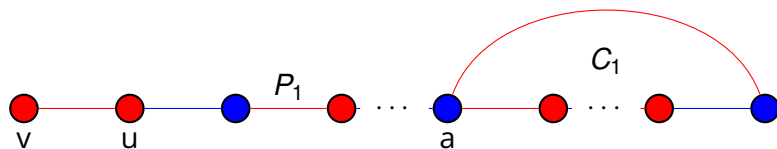
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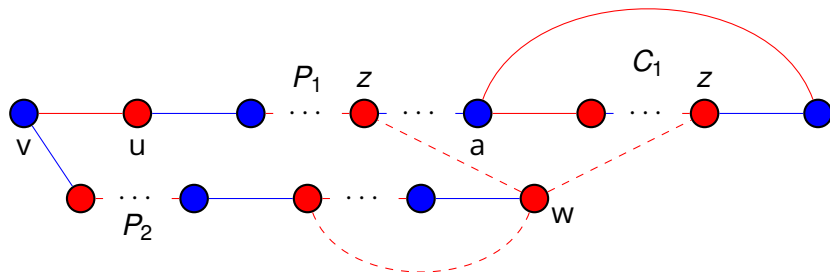


Proof (Continued)



- Reset the vertex coloring.
- Rerun the process starting at v

Proof (Continued)



Alternating Bicycle!



Sharp Threshold Lower Bound

We are now ready to prove:

Theorem

For any constant $\epsilon > 0$ we have that $\mathcal{G}(n, p = \frac{2-\epsilon}{n}, 2)$ is w.h.p adaptably 2-colorable.

Proof.

- Consider an ordered sequence of vertices
 $S = (v_1, v_2, \dots, v_s)$

$$X_S = \begin{cases} 1 & S \text{ forms alternating bicycle} \\ 0 & \text{otherwise} \end{cases}$$

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$$X_S = \begin{cases} 1 & S \text{ forms alternating bicycle} \\ 0 & \text{otherwise} \end{cases}$$

- Whats the probability S forms an alternating bicycle?

$$\mathbb{E}[X_S] = \sum_{2 < i \leq s} \sum_{1 \leq j < s-1} p^{s+1} 2^{-s} = (s-2)^2 p^{s+1} 2^{-s}$$

Proof (Continued)

Recall: $\mathbb{E}[X_S] = (s-2)^2 p^{s+1} 2^{-s} \leq s^2 p^{s+1} 2^{-s}$

- So for $X := \sum_S X_S$ linearity of expectation gives,

$$\mathbb{E}[X] \leq \sum_S \mathbb{E}[X_S] \leq \sum_{s=1}^n n^s s^2 p^{s+1} 2^{-s} = \frac{2(1-\epsilon)}{n} \sum_{s=1}^n s^2 (1-\epsilon)^s$$



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- $\implies \mathbb{E}[X] = \mathcal{O}(n^{-1}) = o(1)$
- Markov's Inequality:
 $\implies \mathbb{P}[X > 0] = \mathbb{P}[X \geq 1] \leq \mathbb{E}[X] = o(1)$



How large can p be?

Recall: $\mathbb{E}[X] \leq \frac{2(1-\epsilon)}{n} \sum_{s=1}^n s^2 (1-\epsilon)^s$

Question

What if $\epsilon = \epsilon(n)$?

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- Let $\epsilon = \lambda_n n^{-1/3}$ where $\lambda_n \rightarrow \infty$ arbitrarily slowly
- $\mathbb{E}[X] \lesssim (1 - o(1)) \frac{2}{n} \sum_{s=1}^n s^2 e^{-s \lambda_n n^{-1/3}}$

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For unimodal sums:

$$\sum_{i=1}^n f(i) = \int_1^n f(x) dx + \mathcal{O} \left(\max_{x \in [1, n]} f(x) \right)$$

How large can p be?

Recall: $\mathbb{E}[X] \leq \frac{2(1-\epsilon)}{n} \sum_{s=1}^n s^2 (1-\epsilon)^s$

Question

What if $\epsilon = \epsilon(n)$?

Want: $\mathbb{E}[X] = o(1)$

- Let $\epsilon = \lambda_n n^{-1/3}$ where $\lambda_n \rightarrow \infty$ arbitrarily slowly
- $\mathbb{E}[X] \lesssim (1 - o(1)) \frac{2}{n} \sum_{s=1}^n s^2 e^{-s \lambda_n n^{-1/3}}$

For unimodal sums:

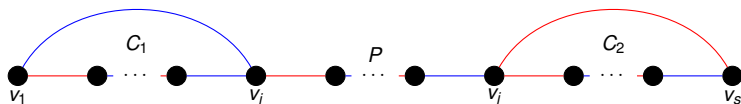
$$\sum_{i=1}^n f(i) = \int_1^n f(x) dx + \mathcal{O}\left(\max_{x \in [1, n]} f(x)\right)$$

So some careful analysis shows,

$$\mathbb{E}[X] \lesssim \mathcal{O}(\lambda_n^{-3}) = o(1)$$

Sharp Threshold Upper Bound

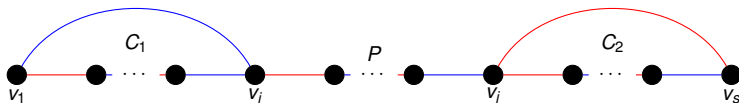
WTS: for $p = \frac{2(1+\epsilon)}{n}$ there exists w.h.p a proper alternating bicycle



Strategy: 2 Steps

Sharp Threshold Upper Bound

WTS: for $p = \frac{2(1+\epsilon)}{n}$ there exists w.h.p a proper alternating bicycle

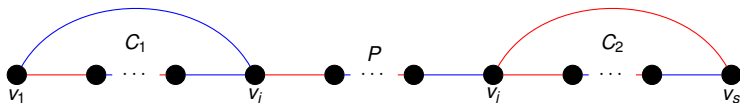


Strategy: 2 Steps

- 1 There exists an alternating path of length $\frac{\epsilon^2 n}{5}$

Sharp Threshold Upper Bound

WTS: for $p = \frac{2(1+\epsilon)}{n}$ there exists w.h.p a proper alternating bicycle



Strategy: 2 Steps

- 1 There exists an alternating path of length $\frac{\epsilon^2 n}{5}$
- 2 Increase p by $\epsilon n^{-4/3}$. Introducing these edges produces a proper alternating bicycle.

Step 1: Alternating path of length $\frac{\epsilon^2 n}{5}$

Lemma

Let $\epsilon > 0$, then w.h.p $\mathcal{G}(n, p = 2(1 + \epsilon)/n, 2)$ contains an alternating path of length at least $\epsilon^2 n/5$.

Follows proof from [Krivelevich, Sudakov '13].

Step 1: Alternating path of length $\frac{\epsilon^2 n}{5}$

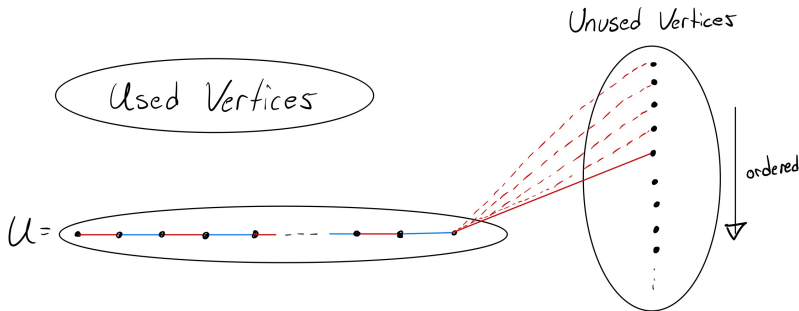
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Sketch.

- Depth-First-Search + Chernoff Bounds



Step 2: Sprinkling

- $p = \frac{2(1+\epsilon)}{n}$

Step 2: Sprinkling

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- $P = v_1 v_2 \cdots v_s$ alternating path $s = \epsilon^2 n/5$.

Step 2: Sprinkling

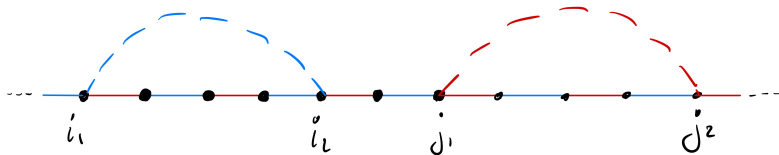
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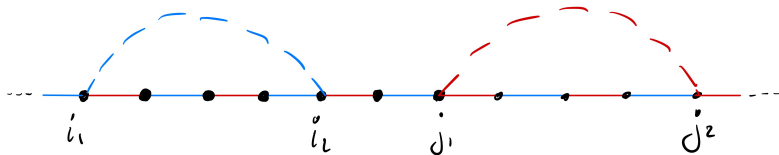
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- Increase p by ϵ/n
- For $S = i_1 < i_2 < j_1 < j_2 \in \binom{I}{4}$



$$X_S = \begin{cases} 1 & S \text{ makes alternating bicycle} \\ 0 & \text{otherwise} \end{cases}, \quad \mathbb{E}[X_S] = \left(\frac{\epsilon}{2n}\right)^2$$

Step 2: Sprinkling (Continued)

Recall: $\mathbb{E}[X_S] = \left(\frac{\epsilon}{2n}\right)^2$

For $X = \sum_{S \in \binom{[n]}{4}} X_S$

$$\mathbb{E}[X] = \sum_{S \in \binom{[n]}{4}} \mathbb{E}[X_S] = \binom{s/2}{4} \left(\frac{\epsilon}{2n}\right)^2$$

Step 2: Sprinkling (Continued)

Recall: $\mathbb{E}[X_S] = \left(\frac{\epsilon}{2n}\right)^2$

For $X = \sum_{S \in \binom{[t]}{4}} X_S$

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If $\mathbb{E}[X^2] = (1 + o(1))\mathbb{E}[X]$ then Chebyshev's Inequality $\implies$
 $X > 0$ w.h.p.

Step 2: Sprinkling (Continued)

Recall: $\mathbb{E}[X_S] = \left(\frac{\epsilon}{2n}\right)^2$

For $X = \sum_{S \in \binom{[l]}{4}} X_S$

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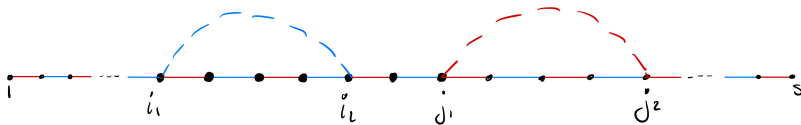
Question

How many $S, S' \in \binom{[l]}{4}$ are there such that X_S and $X_{S'}$ dependent?

Step 2: Sprinkling (Continued)

Question

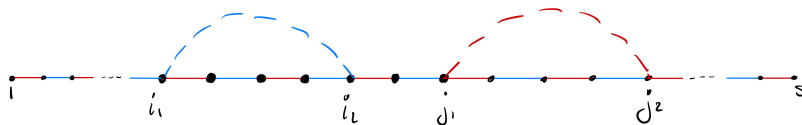
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Step 2: Sprinkling (Continued)

Question

How many $S, S' \in \binom{[l]}{4}$ are there such that X_S and $X_{S'}$ dependent?



$$S = (-, -, -, -)$$

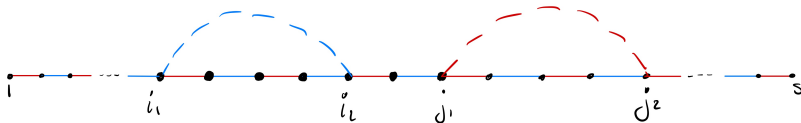
$$S' = (-, -, -, -)$$

$$\# \text{Dependent Pairs} \leq$$

Step 2: Sprinkling (Continued)

Question

How many $S, S' \in \binom{[s]}{4}$ are there such that X_S and $X_{S'}$ dependent?



$$S = (a, b, -, -)$$

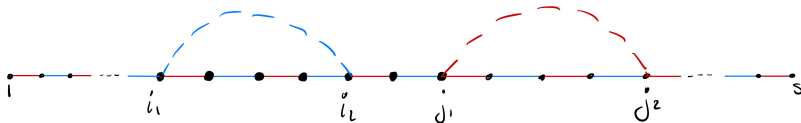
$$S' = (a, b, -, -)$$

$$\# \text{Dependent Pairs} \leq (\# \text{ choices for } a \text{ and } b) \left(\binom{(s-b)/2}{2}^2 + \right)$$

Step 2: Sprinkling (Continued)

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How many $S, S' \in \binom{[s]}{4}$ are there such that X_S and $X_{S'}$ dependent?



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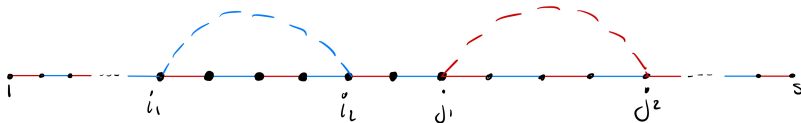
$$S' = (-, -, a, b)$$

$$\# \text{Dependent Pairs} \leq \binom{s/2}{2} \left(\binom{s/2}{2}^2 + \binom{a/2}{2}^2 + \right)$$

Step 2: Sprinkling (Continued)

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How many $S, S' \in \binom{[s]}{4}$ are there such that X_S and $X_{S'}$ dependent?



$$S = (a, b, -, -)$$

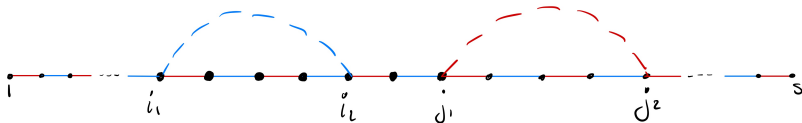
$$S' = (-, -, a, b)$$

$$\# \text{Dependent Pairs} \leq \binom{s/2}{2} \left(2 \binom{s/2}{2}^2 + \binom{(s-b)/2}{2} \binom{a/2}{2} + \right)$$

Step 2: Sprinkling (Continued)

Question

How many $S, S' \in \binom{[s]}{2}$ are there such that X_S and $X_{S'}$ dependent?



$$S = (-, -, a, b)$$

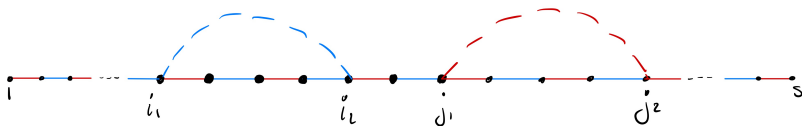
$$S' = (a, b, -, -)$$

$$\# \text{Dependent Pairs} \leq \binom{s/2}{2} \left(3 \binom{s/2}{2}^2 + \binom{a/2}{2} \binom{(s-b)/2}{2} \right)$$

Step 2: Sprinkling (Continued)

Question

How many $S, S' \in \binom{[s]}{4}$ are there such that X_S and $X_{S'}$ dependent?



$$\# \text{ Dependent Pairs} \leq 4 \binom{s/2}{2}^2 = \Theta(s^6)$$

- Small compared to total number of pairs: $\binom{s/2}{4}^2 = \Theta(s^8)$

$$\mathbb{E}[X_S X_{S'}] = \mathbb{E}[X_S] \mathbb{E}[X_{S'}] \left(\frac{2n}{\epsilon} \right)^{|E(S \cap S')|}$$

Step 2: Sprinkling (Continued)

$$\mathbb{E}[X^2] = \sum_{S, S' \in \binom{[s/2]}{4}} \mathbb{E}[X_S] \mathbb{E}[X_{S'}] \left(\frac{2n}{\epsilon}\right)^{|E(S \cap S')|} = \mathbb{E}[X_S]^2 \sum_{S, S' \in \binom{[s/2]}{4}} \left(\frac{2n}{\epsilon}\right)^{|E(S \cap S')|}$$

So we need,

$$\sum_{S, S' \in \binom{[s/2]}{4}} \left(\frac{2n}{\epsilon}\right)^{|E(S \cap S')|} = \binom{s/2}{4}^2 (1 + o(1))$$

Algebra + # *dependent pairs* = $\Theta(s^6) \implies$ Holds for $\epsilon > 0$

How Small can p be?

Lemma

Let $\epsilon = \epsilon(n)$ such that $\epsilon \gg n^{-1/3} \log^{1/3} n$, then w.h.p
 $\mathcal{G}(n, p = 2(1 + \epsilon)/n, 2)$ contains an alternating path of length at
 least $\epsilon^2 n/5$.

For $\lambda_n \rightarrow \infty$ and $\alpha > 0$:

$$\epsilon = \lambda_n n^{-\alpha} \implies |P| \geq \frac{\lambda_n^2 n^{1-2\alpha}}{5}$$

Sadly if $\alpha = 1/3$ then $P \geq \frac{\lambda_n^2 n^{1/3}}{5}$

$$\sum_{S, S' \in \binom{[P]}{4}} \left(\frac{2n}{\epsilon} \right)^{|E(S \cap S')|} \geq (\# \text{ dependent pairs}) \cdot \left(\frac{2n^{4/3}}{\lambda_n} \right)$$

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$$\sum_{S, S' \in \binom{[n]}{s/2}} \left(\frac{2n}{\epsilon} \right)^{|E(S \cap S')|} \geq \Theta(|P|^6) \cdot \left(\frac{2n^{4/3}}{\lambda_n} \right)$$

How Small can p be?

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$$\sum_{S, S' \in \binom{[n]}{s/2}} \left(\frac{2n}{\epsilon} \right)^{|E(S \cap S')|} \geq \Theta(\lambda_n^{12} n^2) \cdot \left(\frac{2n^{4/3}}{\lambda_n} \right)$$

How Small can p be?

Lemma

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Sadly if $\alpha = 1/3$ then $P \geq \frac{\lambda_n^2 n^{1/3}}{5}$

$$\sum_{S, S' \in \binom{[n]}{4^{s/2}}} \left(\frac{2n}{\epsilon} \right)^{|E(S \cap S')|} \geq \Theta(\lambda_n^{11} n^{10/3})$$

How Small can p be?

Lemma

Let $\epsilon = \epsilon(n)$ such that $\epsilon \gg n^{-1/3} \log^{1/3} n$, then w.h.p $\mathcal{G}(n, p = 2(1 + \epsilon)/n, 2)$ contains an alternating path of length at least $\epsilon^2 n/5$.

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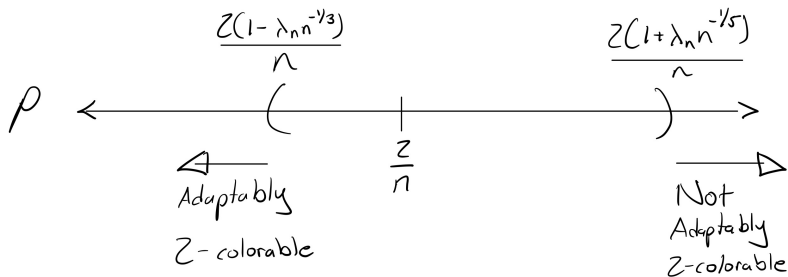
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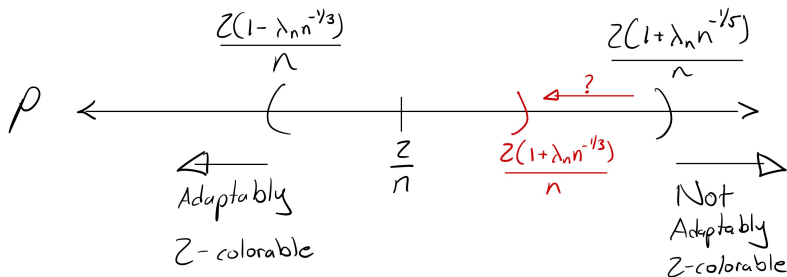
$$\sum_{S, S' \in \binom{[s/2]}{4}} \left(\frac{2n}{\epsilon} \right)^{|E(S \cap S')|} > \Theta(\lambda_n^{16} n^{8/3}) = \left(\frac{s/2}{4} \right)^2$$

So this method doesn't work for $\epsilon = \lambda_n n^{-1/3}$. In fact only works up to $\epsilon = \lambda_n n^{-1/5}$.

Recap

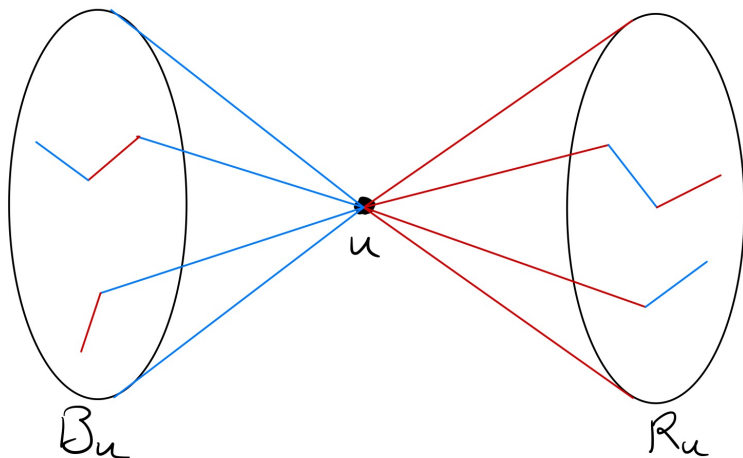


Recap

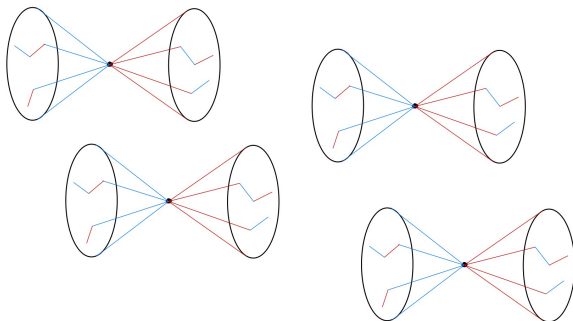


Hourglasses

A technique from [Bollobás, Borgs, Chayes, Kim, Wilson '01]



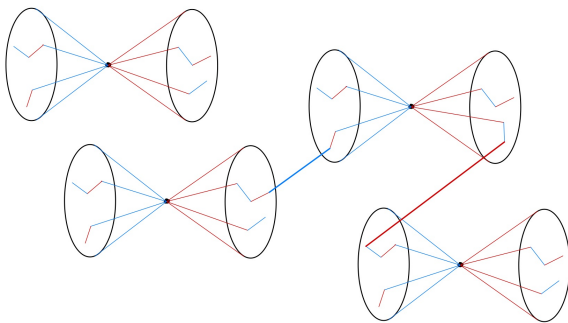
A better Upper Bound



Start with $p = \frac{2(1 - \lambda_n n^{-1/3})}{n}$

There exists $\Theta(\lambda_n^3)$ Hourglasses s.t. $|R_u|, |B_u| \geq \frac{n^{2/3}}{\lambda_n^2}$

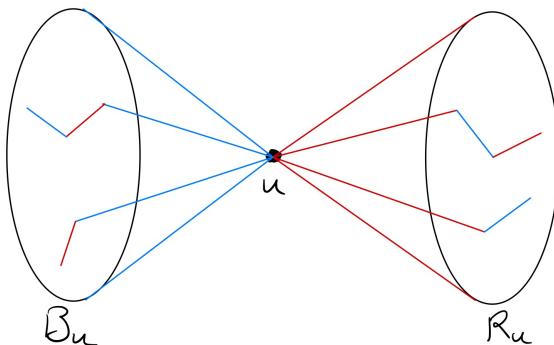
A better Upper Bound



Increase p by $M\lambda_n n^{-4/3}$ for large constant M .

The $\Theta(\lambda_n^3)$ small Hourglasses Merge into a large one, with
 $|R_u|, |B_u| \geq \Theta(\lambda_n n^{2/3})$

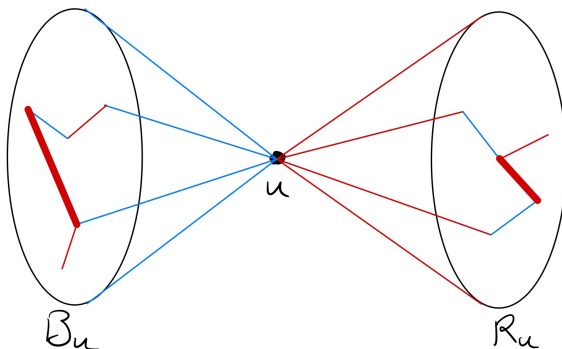
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A better Upper Bound



Increase p by $\lambda_n n^{-4/3}$

Creates alternating odd bicycle!

References

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